

Visualization of Flow Direction using Polarization Angle Changes of Cellulose Nanofiber Suspension

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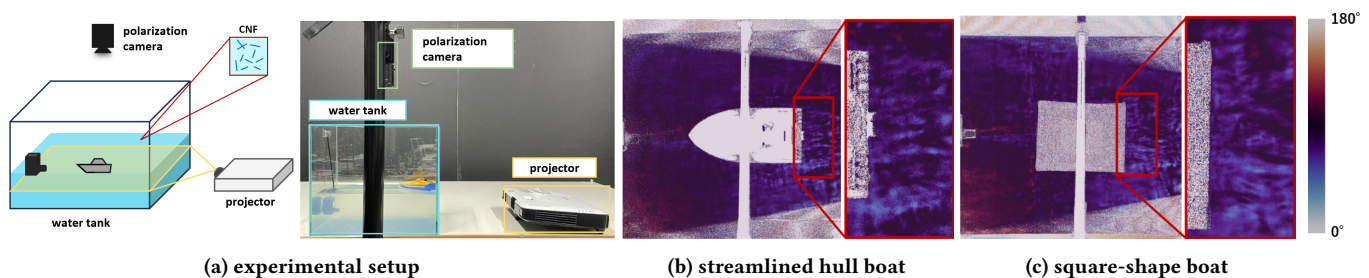


Figure 1: Our method enables the visualization of water flow. CNF is added to water in a glass tank illuminated by a polarized LCD projector, and the flow is captured using a polarization camera (a). To demonstrate the effectiveness of our technique, we placed a streamlined hull and a square-shaped boat in the water flow, as shown in (b) and (c), respectively. A significant difference in the flow patterns, particularly in the wake behind the boats, can be clearly observed.

ABSTRACT

In this study, we propose a novel method for visualizing flow direction by exploiting the polarization angle changes of cellulose nanofiber suspensions. Utilizing a polarization camera, we calculate the polarization angle, enabling pixel-wise visualization of flow direction. We demonstrate that the boat's shape significantly influences the flow patterns generated in its wake. Furthermore, we verify the correspondence between polarization angle and flow direction, and investigate the effects of illumination multiplexing and de-multiplexing on the proposed method. This study presents a promising technique for high-resolution flow direction visualization, with potential applications in fluid dynamics research and industrial settings.

CCS CONCEPTS

• Computing methodologies → Computational photography.

KEYWORDS

flow direction visualization, polarization angle change, cellulose nano fiber

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1 INTRODUCTION

Fluid-motion visualization is essential in diverse fields, including shipbuilding, healthcare, and water sports. Although particle image velocimetry (PIV) is a widely adopted method, its resolution limitations can impede the precise measurement of small-scale flow structures [Raffel et al. 2018]. Yamashita [Yamashita et al. 2019] developed a high-resolution water flow visualization technique using circular polarizers. However, this method is confined to highlighting the existence of flow, without offering insights into the flow direction. Through our observations, we discovered that the polarization angles of cellulose nanofiber (CNF), a type of dietary fiber, vary in response to the flow direction when mixed into a liquid. This paper presents a novel method for visualizing water flow direction by calculating the polarization angle using a polarization camera. In contrast to conventional techniques, our approach

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enables real-time calculation and visualization of the polarization angle with high resolution, offering a significant improvement in flow direction analysis.

2 PROPOSED METHOD

Our key insight is that CNF exhibits a unique property: they align their particles in response to the flow direction, resulting in a rotation of the polarization angle. To visualize the flow direction, we calculate the angle of linear polarization (AoLP) using the polarization information in four directions captured by a polarization camera imaging the CNF suspension. The polarization camera is equipped with four linear polarizers oriented at 0° , 45° , 90° and 135° arranged in a regular pattern, enabling the simultaneous capture of polarization information from all four directions in a single shot. The AoLP can be calculated from the polarization information captured by the polarization camera as following:Eq. (1).

$$AoLP(x, y) = \frac{1}{2} \arctan \frac{I_{45}(x, y) - I_{135}(x, y)}{I_0(x, y) - I_{90}(x, y)}. \quad (1)$$

Finally, the calculated AoLP value from Eq. (1) is visualized using a pseudo-color representation at the pixel level to illustrate the flow direction. As the data obtained by the polarization camera is linearly polarized, a color map cycling every 180° is employed for the color representation. The optical setup for visualizing the flow consists of a polarization camera and a projector as shown in Fig. 1(a). We used EPSON EB-1795F 3LCD projector which illuminates polarized light, and IDS U3-3080CP-P-GL Rev. 2 as a polarization camera. The polarization camera and projector were placed at the top and side of the water tank filled with water containing CNF. In this experiment, we placed two types boats on the water, and water pump in front of the boat to visualize the difference of water flow. Figure 1(b) and (c) shows the water flow around the streamlined hull and square-shape boats, respectively. We are able to recognize larger turbulence behind the square-boat.

To validate the relationship between water flow direction and AoLP, we also captured images of a vortex in a glass cup, as shown in Fig. 2. The vortex exhibited circular rotation along the edge of the glass cup, and the AoLP correspondingly rotated. This clearly demonstrates that the AoLP represents the water flow direction.

Furthermore, it is possible to capture the volumetric water flow by sweeping the projector illumination plane from top to bottom, simultaneously. To achieve higher volumetric resolution, the thickness of the illumination plane must be reduced; however, this results in a lower signal-to-noise (S/N) ratio. In this paper, we also propose an illumination multiplexing and de-multiplexing to improve the S/N ratio. The multiplexing scheme is based on the S-matrix which is an extended Hadamard matrix, and the demultiplexing process is performed using a convex optimization framework with smooth and sparse regularization. Figure 3(b) and (c) show the results of visualization with and without multiplexing, respectively.

3 CONCLUSION

In this study, we demonstrated pixel-wise flow direction visualization using a cellulose nanofiber suspension. The experimental results revealed that the polarization angles varied in response to the water flow direction. To enhance the spatial and temporal resolution of the observation, more efficient observation methods, such as coded sampling, could be applied in future work.

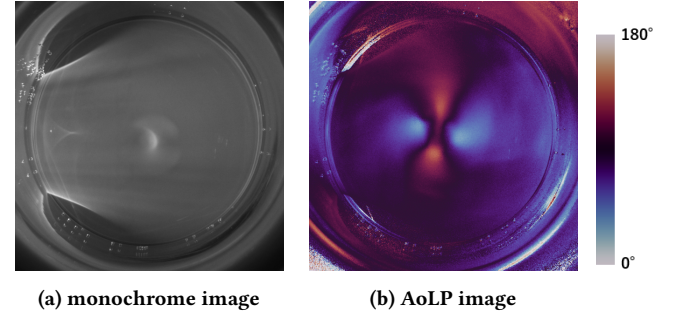


Figure 2: Visualization of vortex water (top view). Water flow is not visible in monochrome camera (a), however, we are possible to visualize the flow through AoLP captured by polarization camera.

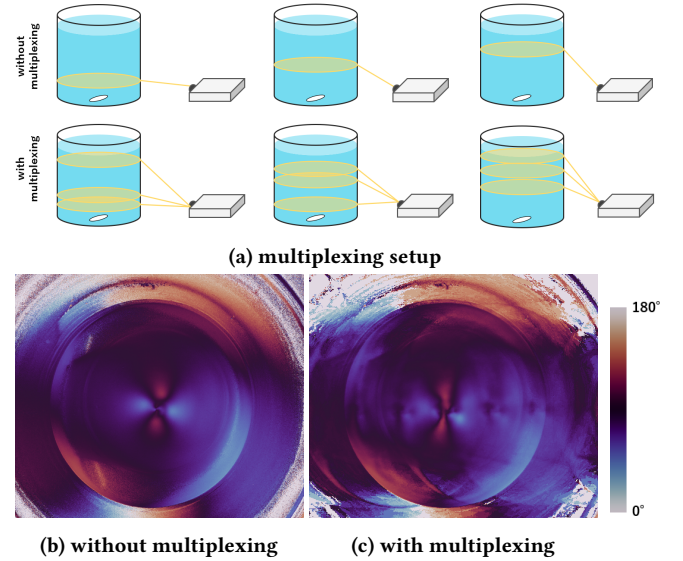


Figure 3: Illumination multiplexing setup and visualization results. (a) shows the illumination multiplexing setup. (b) shows the result of visualization without multiplexing and (c) shows the result of visualization with multiplexing.

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